Econometrics Model with Panel Data

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1. Definition

- Consumption and Income of 5 households in 3 years

Obs	Id	year	Consumption	Income
1	1	2007	1.200	2.000
2	1	2008	1.400	2.100
3	1	2009	1.500	2.200
--	..	..	.	.
13	5	2007	2.000	3.000
14	5	2008	2.100	3.500
15	5	2009	2.300	3.400

1. Definition

- Panel data: the same groups of observation (N) (household, enterprise, individuals, countries…) are observed over time (T)
- The panel data can have:
  - Variable has different values for each obs, but does not change overtime (location, gender..)
  - Variable has different values for each obs, but change overtime (exchange rate, FDI, consumption, income..)

2. Advantages of Panel Data

- For example: (Cross-section dataset) Analyze the relationship between quantity of fertilize and productivity
- The sample is of 20 fields (thửa ruộng)

$$\text{Productivity}\hat{} = 4 + 0.5\text{Fertilize}$$

- Q: Are there any problems with the estimated result?
- A: Yes
  - The productivity also depends on the rich (độ phì nhiêu) of the land that cannot be observed ➔ the model cannot evaluate the effects of the **unobserved variables (u)**.
  - If the variable "the rich of the land" has correlation with the independent variable "Fertilize" ➔ We have the problem of the **correlation between X and u**

2. Advantages of Panel Data

– If 20 fields distributed in 20 different provinces, we want to observe the effect of province ➔ generate 19 dummy variables ➔ too complicated

- **Some conclusions:**

– There are missing unobserved variables (u) and u correlate with X

– These variables are different for each unit

=> Biased OLS estimation

2. Advantages of Panel Data

- Overcome the problem of missing unobserved variables
- The data (units) are observed overtime, so the sample is larger and we can track all the changes of units overtime.

3. Econometric Model for Panel Data

The unobserved variables can be:

- Change/ unchange by unit (i)
- Change/ unchange by time (t)
- Change by both unit (i) and time (t)

3. Econometric Model for Panel Data

- We have model:

$$Y_{it} = \beta_0 + \delta_0 d2_t + \beta_1 X_{it} + a_i + u_{it} \quad t=1,2$$

u_{it} satisfy all the assumptions of the OLS

a_i: unobserved and unchanged overtime (ie. geography location of provinces)

➔ **Fixed effects model (Mô hình này được gọi là mô hình ảnh hưởng cố định)**

3. Econometric Model for Panel Data

- Let: $v_{it} = a_i + u_{it}$
- Depends on the characteristics of a_i, we have 3 models:
  - Pooled OLS (Mô hình hồi qui gộp - POLS)
  - Fixed effect (Mô hình tác động cố định - FE)
  - Random effect (Mô hình tác động ngẫu nhiên-RE)

Pooled OLS - POLS

1. **If there is no existence of a_i:**

- Apply the OLS method. The panel data is considered as pooled cross section data ➔ POLS
- OLS is the best method because error term v satisfy all the assumption of the OLS)

Random Effect Model

2. **If there is the existence of a_i:**

2.1 If a_i does not correlate with X => $v_{it} = a_i + u_{it}$ (composite error term) has no correlation with X

$$Cov(a_i, X) = 0$$

➔RE - random effects model

Random Effect Model

- We have model:

  $Y_{it} = \beta_0 + \beta_1 X_{1it} + .. + \beta_k X_{kit} + a_i + u_{it}$,

- $v_{ij} = a_i + u_{ij:}$ composite error

$\Rightarrow Y_{it} = \beta_0 + \beta_1 X_{1it} + .. + \beta_k X_{kit} + v_{ij}$

$\Rightarrow$RE analyze the effect of a_i on the change of Y We have:

$Var(vij) = \sigma a^2 + \sigma u^2$ ➔ homoskedasticity

Random Effect Model

$cov(v_{ij}; v_{is}) = cov(a_i + u_{ij;}\ a_i + u_{is}) = var(a_i) + cov(a_i, u_{is}) + cov(a_i; u_{ij})$ when $j \neq s$

$cov(v_{ij}; v_{is}) = var(a_i) + var(u_{ij}) + cov(a_i, u_{is}) + cov(a_i; u_{ij})$ khi $j = s$

- These factors are different from 0, even when a and u do not correlate

➔ There is auto-correlation in u

- Estimation Method: GLS for pooled cross-section data

Fixed Effect Model - FE

2.2 When a_i correlate with X, both RE and POLS are biased

$$Cov(a_i, X) \neq 0$$

➔ Cannot group a_i into v_{it}

➔Use FE – fixed effects model with vertical estimation method

** Note: if X is unchanged over time, we can not use FE model.

FE- vertical estimation

$$Y_{it} = \beta_0 + \beta_1 X_{it} + a_i + u_{it}$$

$$\overline{Y}_{it} = \beta_0 + \beta_1 \overline{X}_{it} + a_i + \overline{u}_{it}$$

$$Y_{it} - \overline{Y}_{it} = \beta_1 \left(X_{it} - \overline{X}_{it} \right) + \left(u_{it} - \overline{u}_{it} \right)$$

➔Can delete the effect of unobserved variable ai

FE vs. RE

FE:

- Cannot evaluate the effect of unchanged-overtime variable: gender, capability, initial characteristics

RE

- Assumption about the un-correlation between a and X are too strong.

How to choose the model?

- Nếu biến bị bỏ sót là không đáng kể => MH hồi qui gộp là tốt nhất
- Nếu biến bị bỏ sót không tương quan với X => RE là hiệu quả hơn FE (nhưng phải giả thiết về sự không tương quan giữa a và u)
- Nếu biến bị bỏ sót là tương quan với X thì RE là chệch và không vững=> chọn FE
- Lựa chọn giữa POLS và RE: sử dụng xttest0
- Nếu RE được lựa chọn => sẽ chọn giữa FE hay RE: Hausman

Câu lệnh trong STATA

Bước 1: cần test để xem nên chọn RE/FE hoặc POLS.

➔ Breusch-Pagan test for the significant difference across units.

H0: no significant difference across units (no panel effect) [không tồn tại a_i)

H1: significant difference across units (panel effect) [có tồn tại a_i)

```
xtset [panel variable] [time variable]
Xtreg Y X, re
Xttest0
```

p-value < 0.05 ➔ bác bỏ H0, chấp nhận H1 ➔ RE/FE

p-value >0.05 ➔ chấp nhận H0 ➔ POLS

Câu lệnh trong STATA

Nếu trong Bước 1, H0 bị bác bỏ ➔ HAUSMAN TEST để chọn FE or RE

H0: $Cov(a_i,x_{it}) = 0$

H1: $Cov(a_i,x_{it}) \neq 0$

```
xtreg Y X, fe
est store fe
xtreg Y X, re
hausman fe
```

- **Nếu p-value < 0.05 ➔ bác bỏ H0 ➔ chọn FE**
- **Nếu p-value > 0.05 ➔ chấp nhận H0 ➔ choose RE**